

Deren Lei

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EDUCATION

University of California, Santa Barbara (UCSB)

Computer Science B.S.

Planning to Declare: Minor in Statistical Science

Program: College of Engineering Honors Program & College of Letter and Science Honors Program

Major GPA: **3.92** (Around Top 2.5%)

SANTA BARBARA, CA

Fall 2015 – Spring 2019

RESEARCH EXPERIENCE

UCSB Natural Language Processing Lab

Researcher (Advisor: William Wang)

SANTA BARBARA, CA

Fall 2017 – Present

- **Multimodal headline generation (Work in progress)**

Currently doing data collection from CNN/Daily Mail: headline, story highlights, article, images. Will submit this year.

- **Implicit regularization of stochastic gradient descent in NLP (First author, in submission)**

Studied the implicit regularization of stochastic gradient descent in practical deep learning systems and its correlation with explicit techniques. Results can directly contribute to future NLP studies by helping researchers measure models performances better.

- **Natural language instruction learning for intelligent agents**

Assist PhD researchers for their engineering problems when get into NLP area. Solved open-source project issues. Did model hyperparameter optimization. Simultaneously learnt deep learning model implementation via Tensorflow and PyTorch and participated in weekly NLP group reading.

UCSB Dynamic Networks: Analysis and Modeling (Dynamo) Lab

Research Assistant (Advisor: Ambuj Singh)

SANTA BARBARA, CA

Fall 2016 – Winter 2016

- **Human Connectom Project, WU-Minn-Oxford consortium**

To get hands-on of data mining when I just started my computer science study. Participate in weekly meeting, learnt the skill of implementing machine learning and feature engineering. Using MATLAB and python to do data preprocessing. Help me make decision for continue exploring machine learning related area at UCSB.

WORK EXPERIENCE

Amazon

BEIJING, CHINA

Software Engineering Intern (Mentor: Gang Wang)

Summer 2017

- Created an internal tool to help developers to do automatic product page status checking. Saved them more than 5 hours of manual checking the special offer pages per month.
- Did more than 100000 data processing to test a softline product recommendation deep learning algorithm that could applied to Amazon product recommendation system in recent years.

PROJECTS

Activity Recognition (Advisor: B.S. Manjunath)

Winter 2018

- Helping researchers in UCSB Vision Research Lab measuring Residual Computer Networks' performances by learning via single image frames on AVA dataset. The results can help their project of analyzing people's behavior in hospital.

NoteHub (Developer, Product Manager)

Winter 2018

- An software system with android-end and web-end (mainly focused) for students and professors to share notes in both hand-written and electrical format using Java and Javascript.
- Contact Scanbot Team to freely use their API for education purpose when it costs €100,000.

GaucheMap (Group Leader)

Spring 2017

- Developed a map based event sharing system on android platform using Google Maps APIs. To help UCSB students involved in social activities. Get hands-on android development experience with seniors while learning basic engineering skills with other sophomores.

SKILLS

Programming Languages: Python, Java, C++, Scala, MATLAB, R, MIPS Assembly, Javascript

Technical Tools: Tensorflow, PyTorch, Git, AWS, LaTeX, SQL, Firebase